

Experience

Lucid Motors

Senior Software Engineer (Maps and Navigation)

Newark, CA

Nov 2023 – Present

- Led development of HD map software on Nvidia Xavier/Orin with QNX, integrating GNSS sensor data with TomTom APIs.
- Delivered ADAS features (Lane Change Assist, Adaptive Curve Speed Control, Hands Free Highway Assist) through cross-functional collaboration with platform, perception, motion planning, cloud, and field testing teams.
- Enhanced system stability and runtime reliability through memory, CPU, and bandwidth profiling, improving offline performance and reducing error rates.
- Created interactive visualization and debugging tools with Folium, Plotly, and QGIS, enhancing map data analysis and accelerating issue resolution in ADAS development.

Dematic, Kion Mobile Automation

Machine Learning Engineer (Perception, Localization, and Mapping)

Holland, MI

Sept 2020 – Oct 2023

- Developed on-board mapping and localization software for autonomous guided vehicles (AGVs) in warehouse environments.
- Achieved sub-centimeter localization accuracy by refining reflector and line-extraction algorithms.
- Accelerated localization pipeline by 30% through optimizing feature-based association algorithms.
- Reduced technical debt in the mapping codebase by simplifying APIs, decoupling modules, and introducing caching threads; enhanced FastSLAM implementation for robust real-time performance.
- Developed YOLOv3-based perception model for real-time detection of persons, forklifts, and pallets using 3D camera data.

Stony Brook University

Research Assistant

Stony Brook, NY

May 2017 – Aug 2020

- Developed machine-learning and algebraic algorithms for simulating and synthesizing robotic mechanisms, leading to multiple peer-reviewed ASME publications.
- Built [MotionGen](#), a web-based mechanism design framework using the MEAN stack, delivering a RESTful service with cross-platform iOS/Android apps via Apache Cordova.

Education

Stony Brook University

Ph.D., Mechanical (Concentration: Design and Robotics, Minor: Applied Mathematics), GPA 3.95

Stony Brook, NY

Aug 2015 – Aug 2020

- **Relevant Courses:** Robotics, Dynamics, Control, Kinematics, Stress Analysis, Optimization, Algorithms

Selected Projects

Mapping & Localization

- Developed SLAM pipelines (gmapping, RTAB-Map) with LiDAR and 3D cameras, generating occupancy grids and loop-closed maps for robust AGV/AMR localization.
- Implemented particle filter and extended Kalman filter for multi-sensor fusion (GPS, LiDAR, RADAR), reducing localization drift under noisy sensor conditions and improving reliability of autonomous navigation pipelines.

Perception & Machine Learning

- Built a stereo-camera distance estimator for front-view vehicles. Used SGBM disparity calculation for depth and YOLO for detection, enabling real-time ADAS perception.
- Implemented 3D reconstruction with structure-from-motion (SfM). Applied SIFT feature matching, computed fundamental and essential matrices, aligned image sequences, and generated point-cloud models from monocular data.
- Developed a 3D object-detection framework combining camera and LiDAR. Applied sensor fusion to project point clouds into images, detect objects, and estimate 3D bounding boxes, improving perception accuracy in autonomous driving.
- Built a traffic-sign classification pipeline using CNNs trained on 40+ classes in Keras, achieving over 95% accuracy.
- Created real-time lane detection pipeline using BEV transforms and polynomial fitting.

Planning & Control

- Designed an FSM-based trajectory planner and PID controller for autonomous highway driving; integrated perception, localization, planning, and control modules within ROS to achieve end-to-end autonomy.
- Implemented autonomous pick-and-drop operations in Gazebo simulations by combining AMCL-based localization with path planning algorithms, demonstrating full autonomy in simulated warehouse workflows.

Technical Proficiency

- **Programming:** C++, Python, JavaScript, Matlab, Mathematica, Delphi
- **Machine Learning & Robotics:** TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, Pandas, Open3D, CNNs
- **Mapping & Localization:** ROS, Gazebo, Rviz, gmapping, RTAB-Map, AMCL, EKF, particle filter, QGIS
- **Tools & Infrastructure:** Git, Anaconda, Jupyter, VirtualBox, Optimization(LTtng, Perf, Valgrind)

Selected Publications

- Sharma S., Purwar A.; **A Machine Learning Approach to Solve the Alt–Burmester Problem for Synthesis of Defect-Free Spatial Mechanisms.** ASME J. Computing and Information Science in Engineering; doi:10.1115/1.4051913
- Sharma S., Purwar A.; **Path Synthesis of Defect-Free Spatial 5-SS Mechanisms Using Machine Learning.**, ASME IDETC-CIE2020; doi:10.1115/DETC2020-22731
- Sharma S., Purwar A., Ge Q.J.; **Motion Synthesis Approach to Solving Alt-Burmester Problem by Exploiting Fourier Descriptor Relationship Between Path and Orientation.**, ASME J. Mechanisms Robotics; doi:10.1115/1.4042054